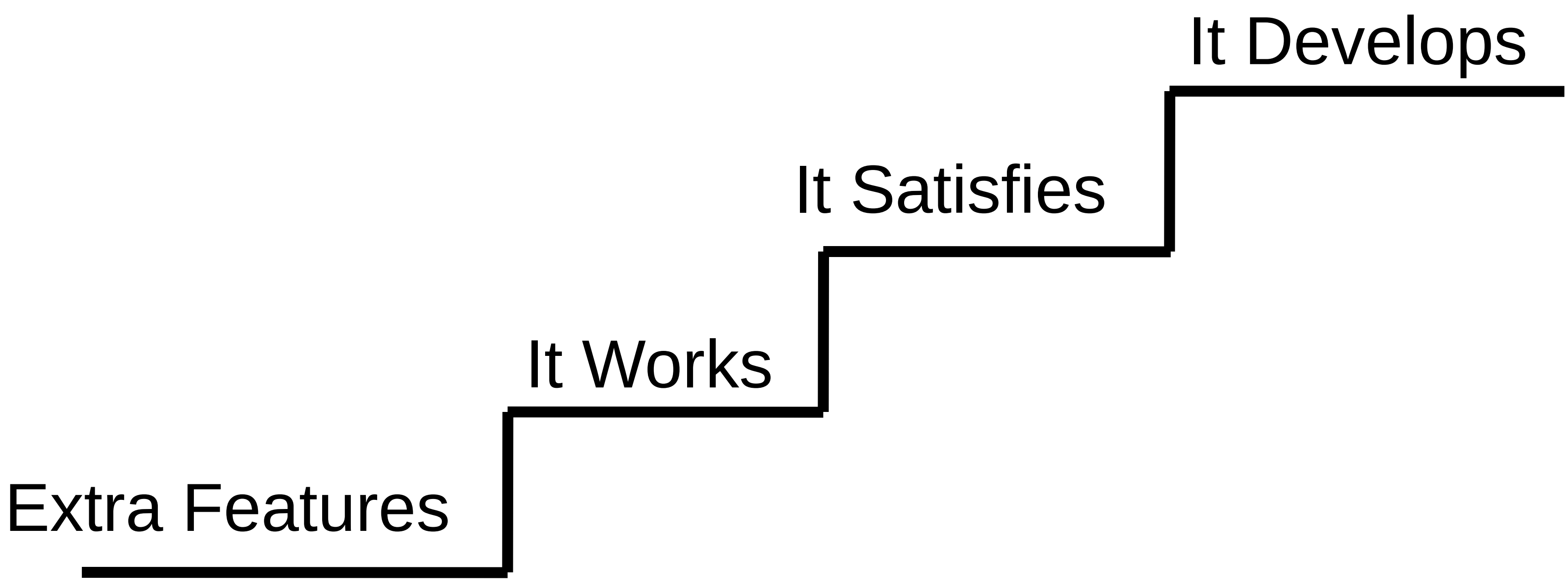
QUALITY MANAGEMENT - IT'S ROLE IN CONSTRUCTION COST EFFECTIVENESS

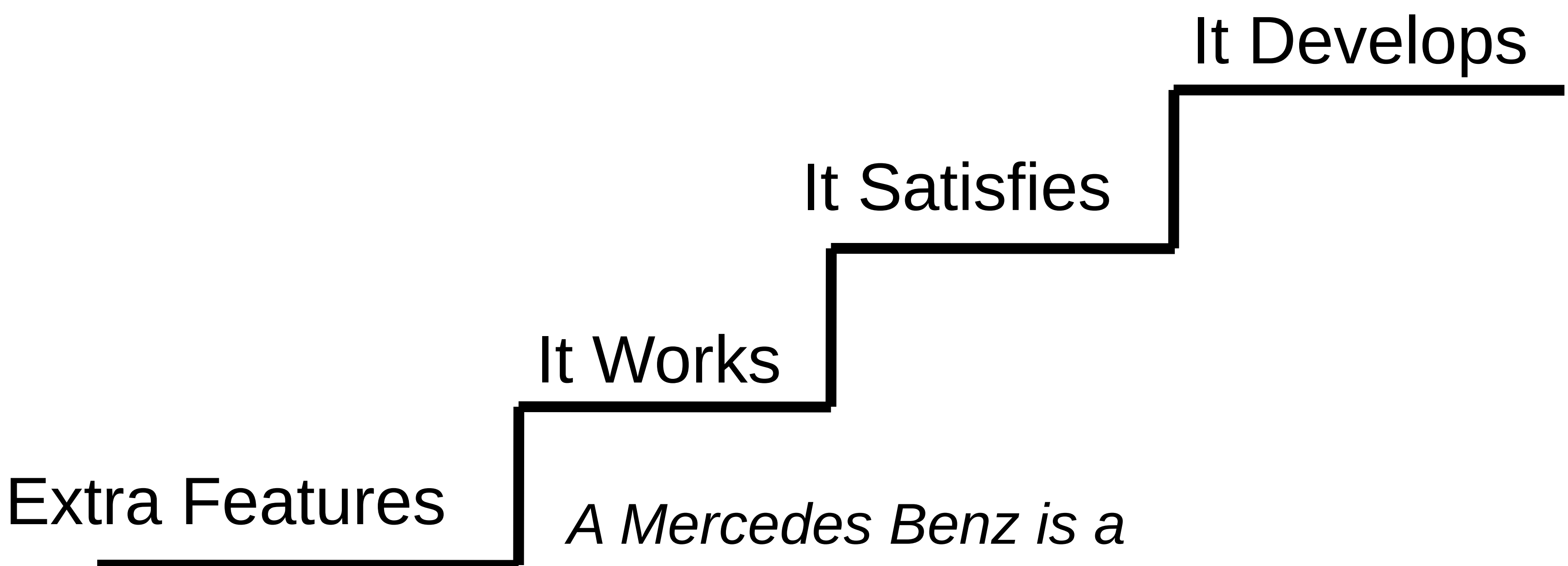
Dr. K. N. Satyanarayana
Building Technology & Construction Management Division
Department of Civil Engineering
Indian Institute of Technology, Madras
satyakn@iitm.ac.in

HIERARCHY OF QUALITY DEFINITIONS



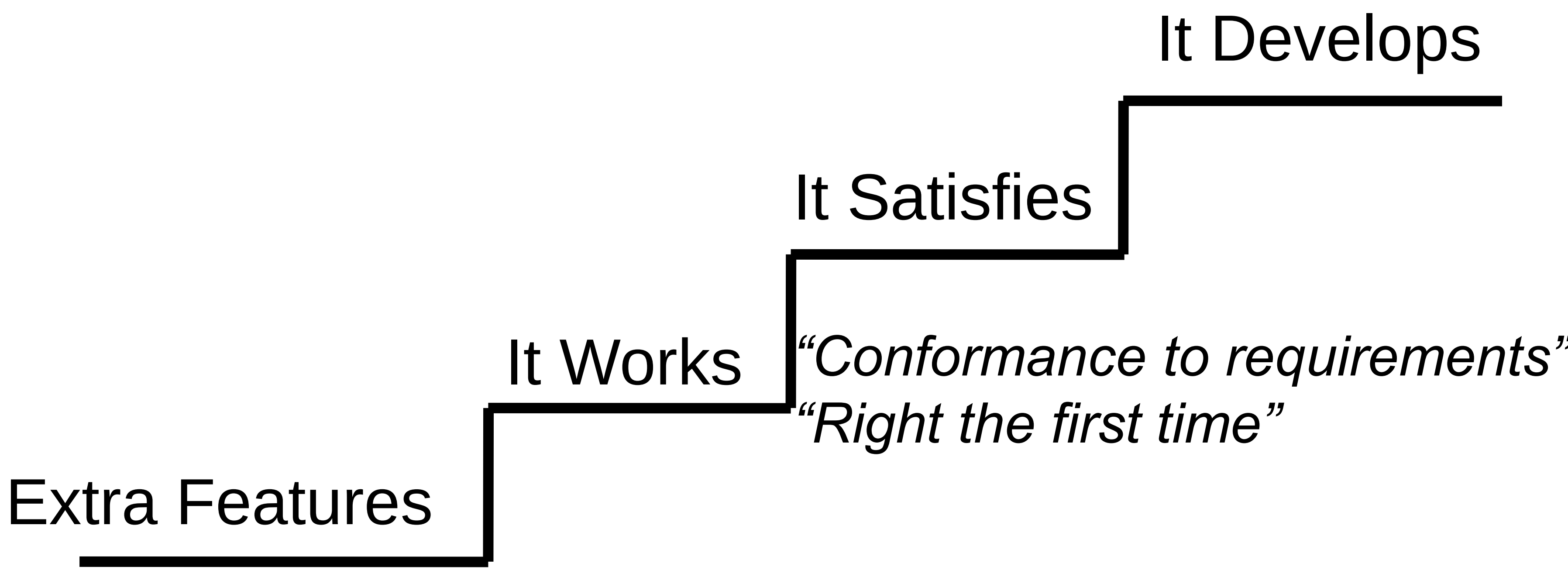
Ronald Dumas

HIERARCHY OF QUALITY DEFINITIONS



A Mercedes Benz is a better quality car than a Maruthi 800

HIERARCHY OF QUALITY DEFINITIONS



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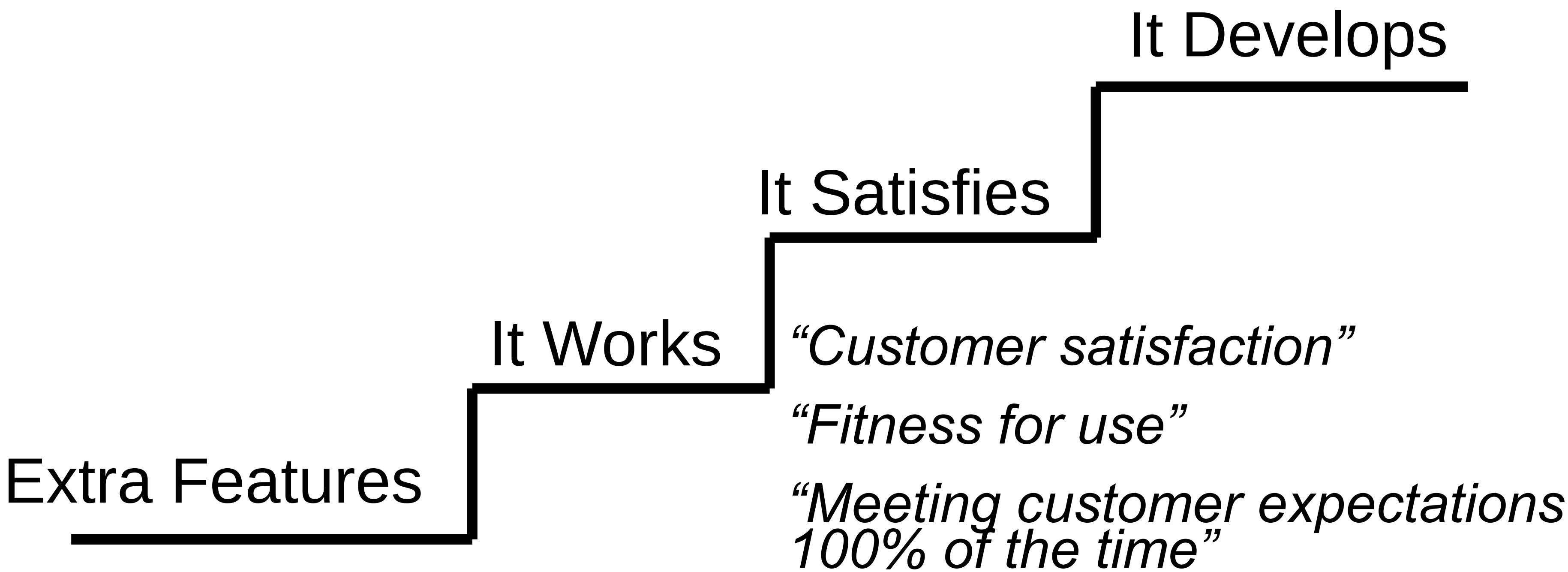
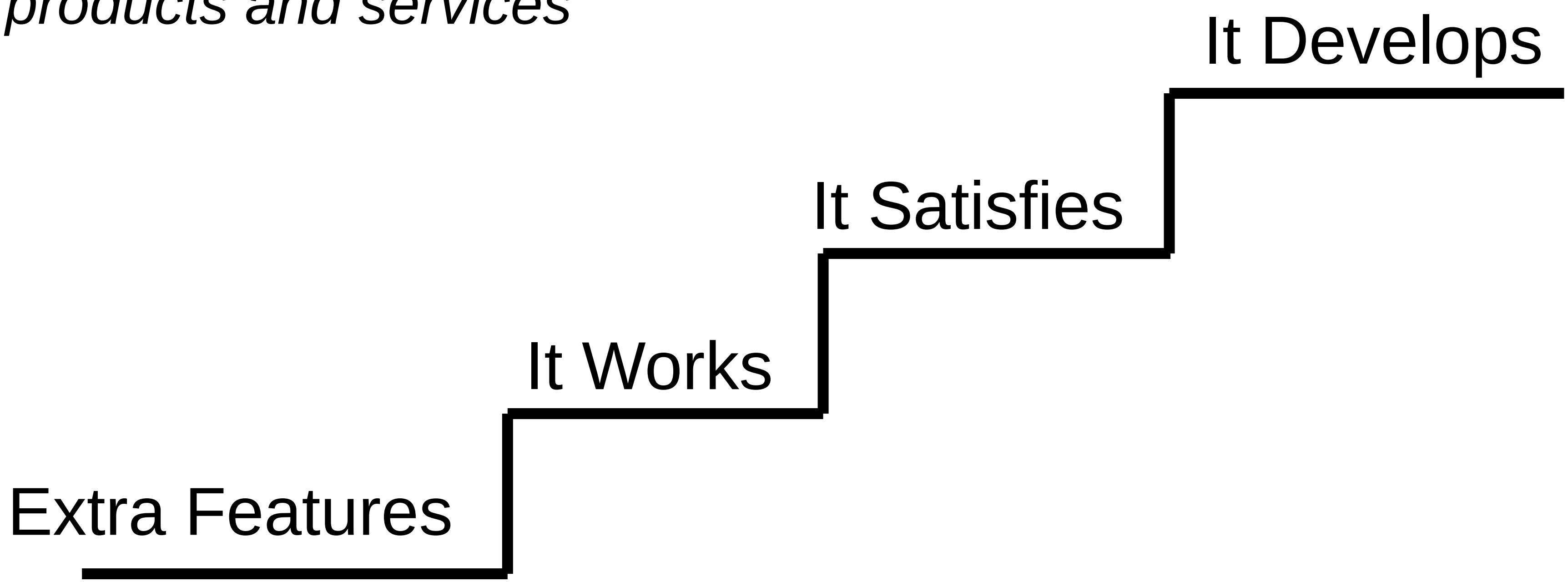


HIERARCHY OF QUALITY DEFINITIONS

Develop customers by educating them and exposing them to greater value from products and services



QUALITY FUNCTIONS - DEFINITIONS

- Quality Management

  - Concerns all the activities in producing a quality product. As such, it includes the management of the specific functions of quality assurance, quality control, and quality audit.

- Quality Assurance

  - Those planned, systematic and preventive actions which are required to ensure that materials, products, or services will meet specified requirements.

QUALITY FUNCTIONS - DEFINITIONS

- Quality Control

  - Inspection, test, or examination to ensure that materials, products, or services will meet specified requirements.

- Quality Audit

  - A documented investigation conducted by the manufacturer to verify that applicable requirements are being implemented.

TOTAL QUALITY MANAGEMENT

- A complete management philosophy that permeates every aspect of a company and places quality as a strategic issue.

- It is accomplished through an integrated effort among all levels of a company to increase ***customer satisfaction*** by ***continuously improving*** current performance.

DEVELOPMENT OF TQM CONCEPTS

1950s - Japanese introduced to quality concepts

1960s - Japanese started developing concepts of total quality

1970s - Japanese quality products in market

1980s - US manufacturing industry, Europe, Pacific basin

1990s - US construction industry, Indian manufacturing

2000s - Indian construction industry (?)

"…*You will hear experts saying that better quality in Japan comes from their culture. But 35 years ago they had the same culture and made junk.*" - Dr. J. M. Juran

QM Gurus : Deming, Juran, Crosby, Ishikawa, Feigenbaum, …

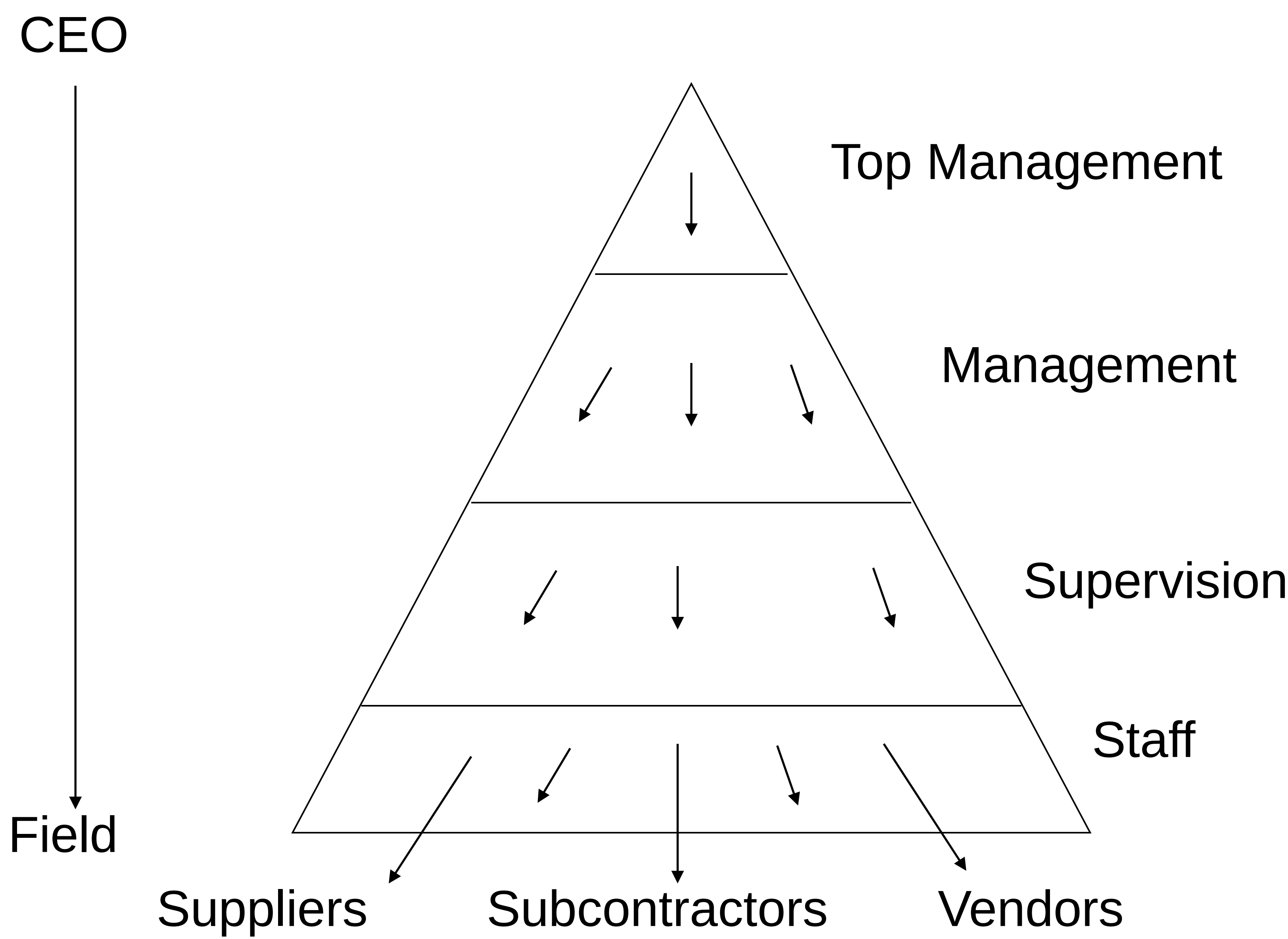
No. of consultants

GUIDING PRINCIPLES OF TQM

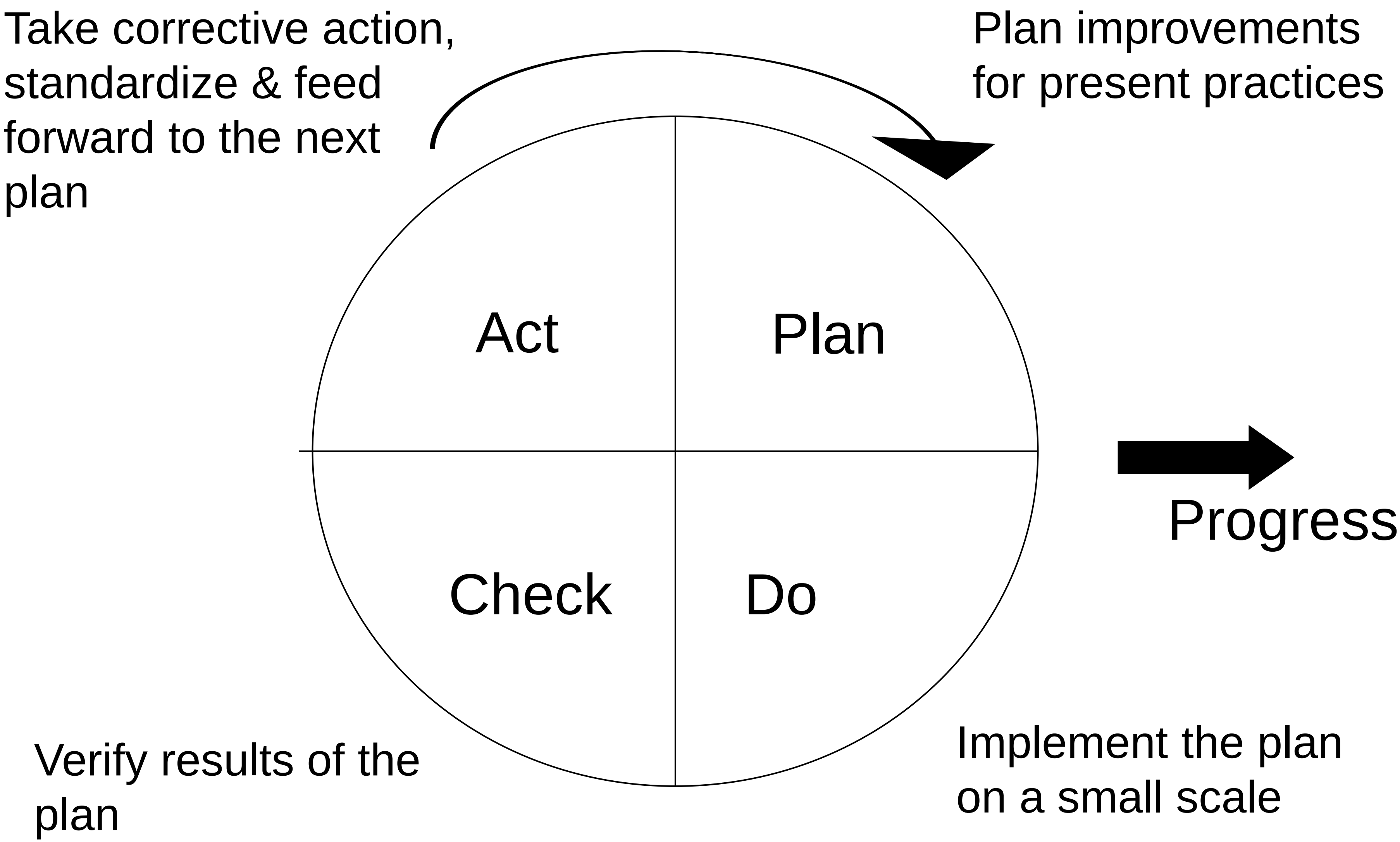
Dr. Steve Smith

- Approach — Management led
- Scope — Company wide
- Scale — Everyone responsible
- Philosophy — Prevention vs. Detection
- Standard — Right first time
- Measure — Cost of quality, other measures
- Theme — Continuous improvement

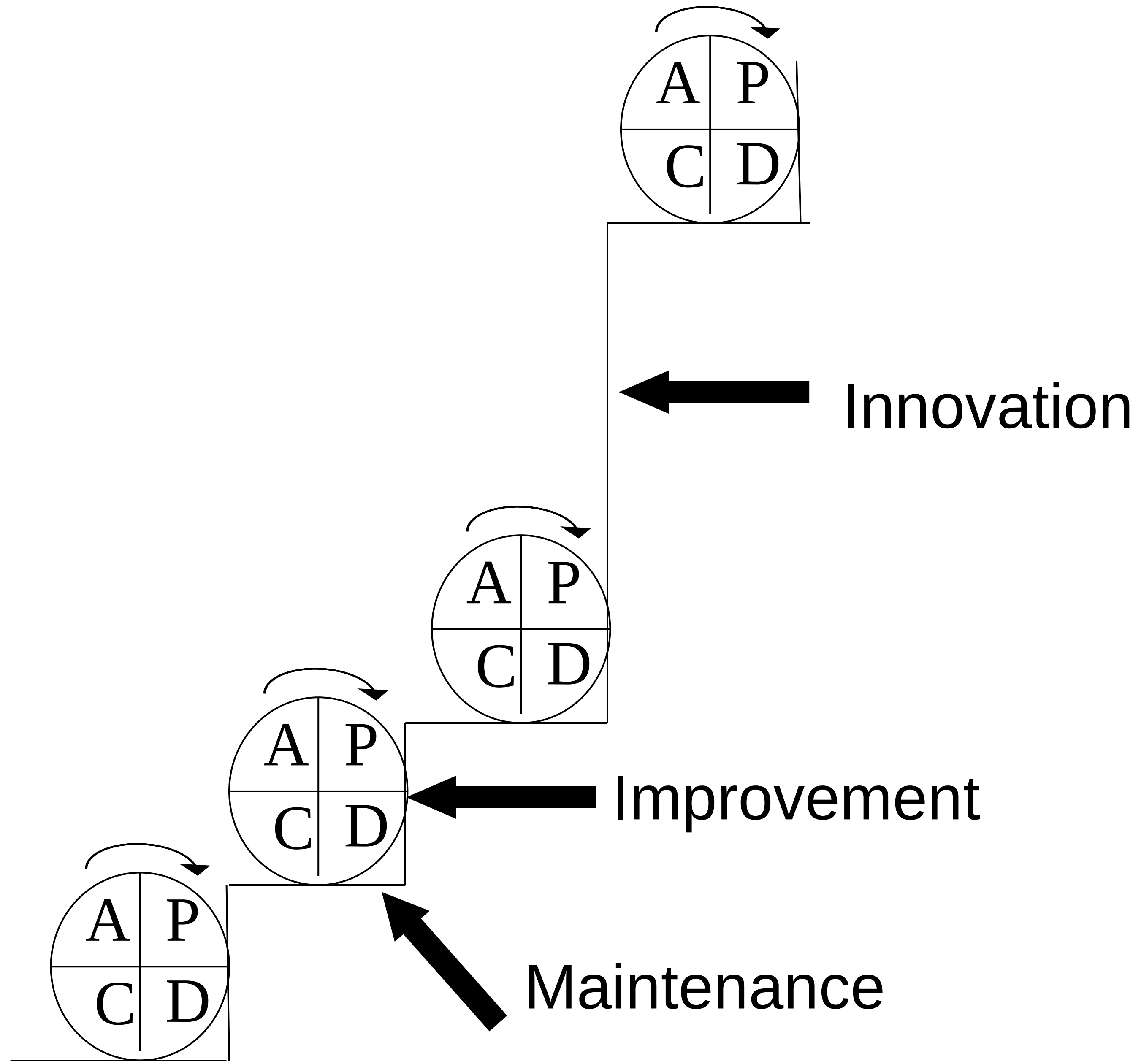
THE TQM CASCADE



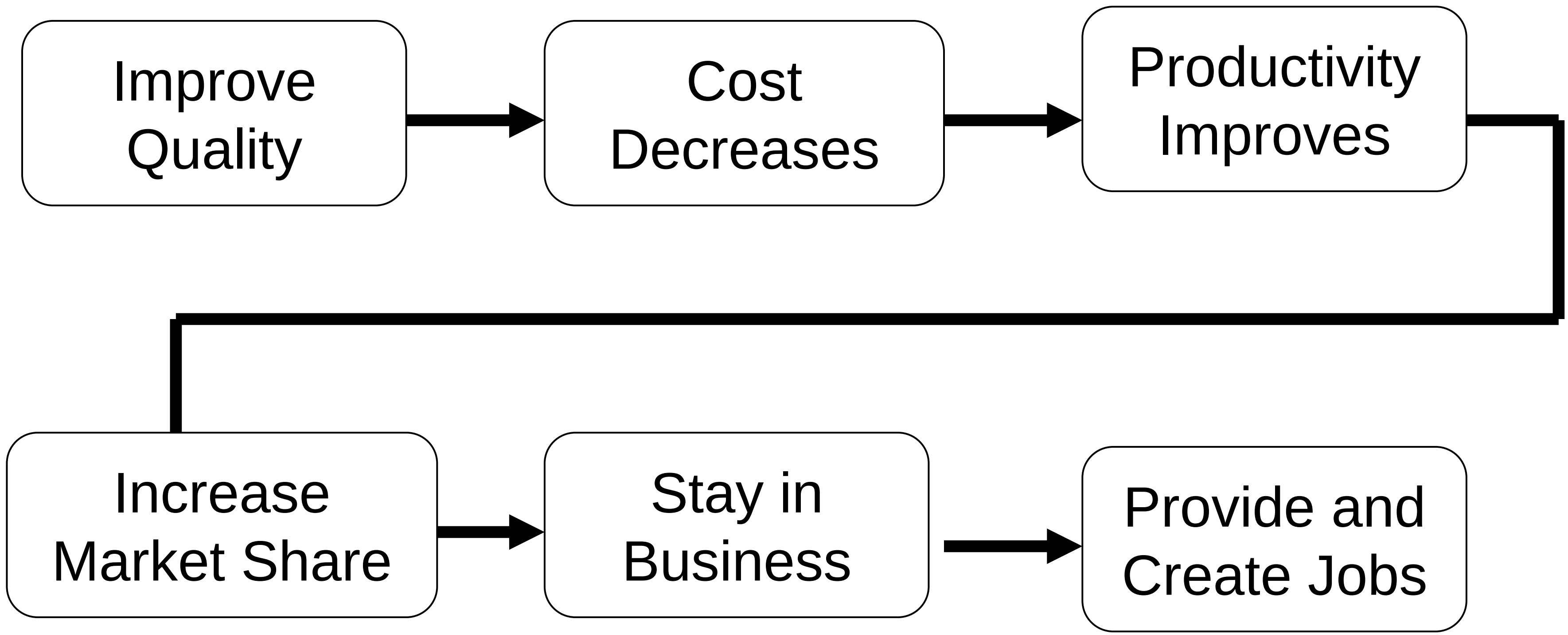
PDCA CYCLE


Take corrective action, standardize & feed forward to the next plan
Plan improvements for present practices
Act
Plan
Check
Do
Progress
Verify results of the plan
Implement the plan on a small scale

MAINTENANCE, IMPROVEMENT & INNOVATION



DEMING CHAIN REACTION

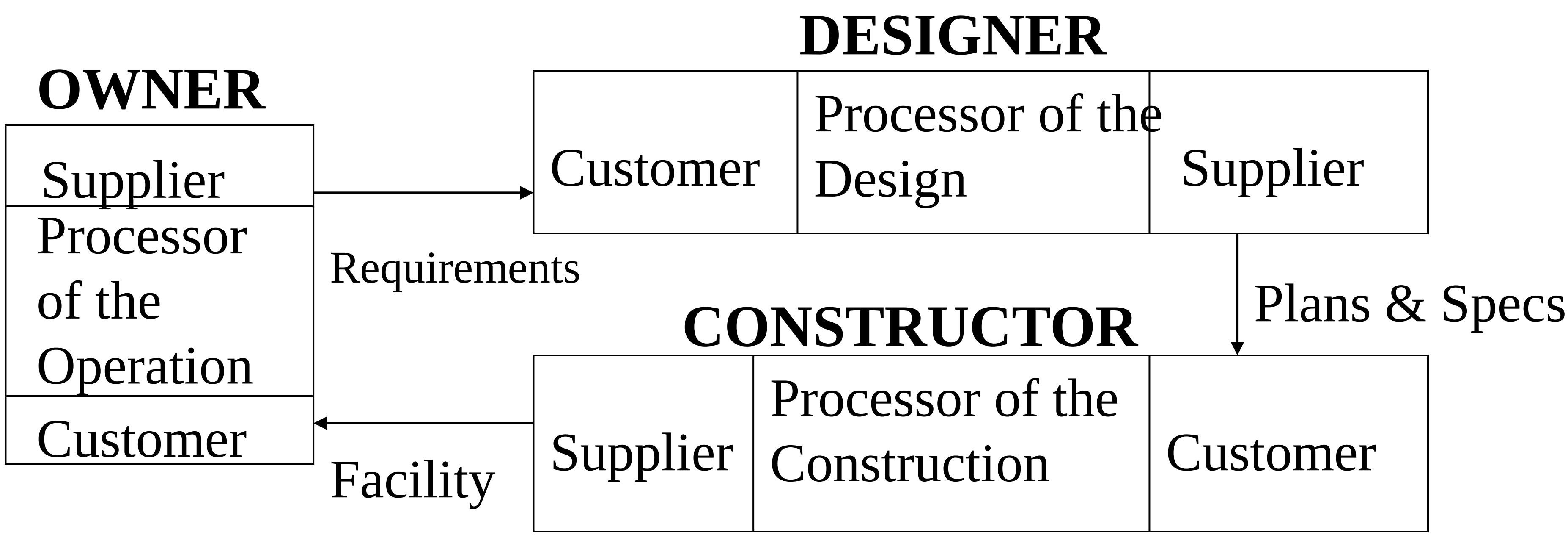


TRIPLE ROLE CONCEPT

- Customer
- Processor
- Supplier

"Our activities must be customer driven. All of our people must understand who their customers are and what their customers expect, regardless of whether those customers are internal or external"

- John Creedon, President, Metropolitan Life



ELEMENTS OF TQM

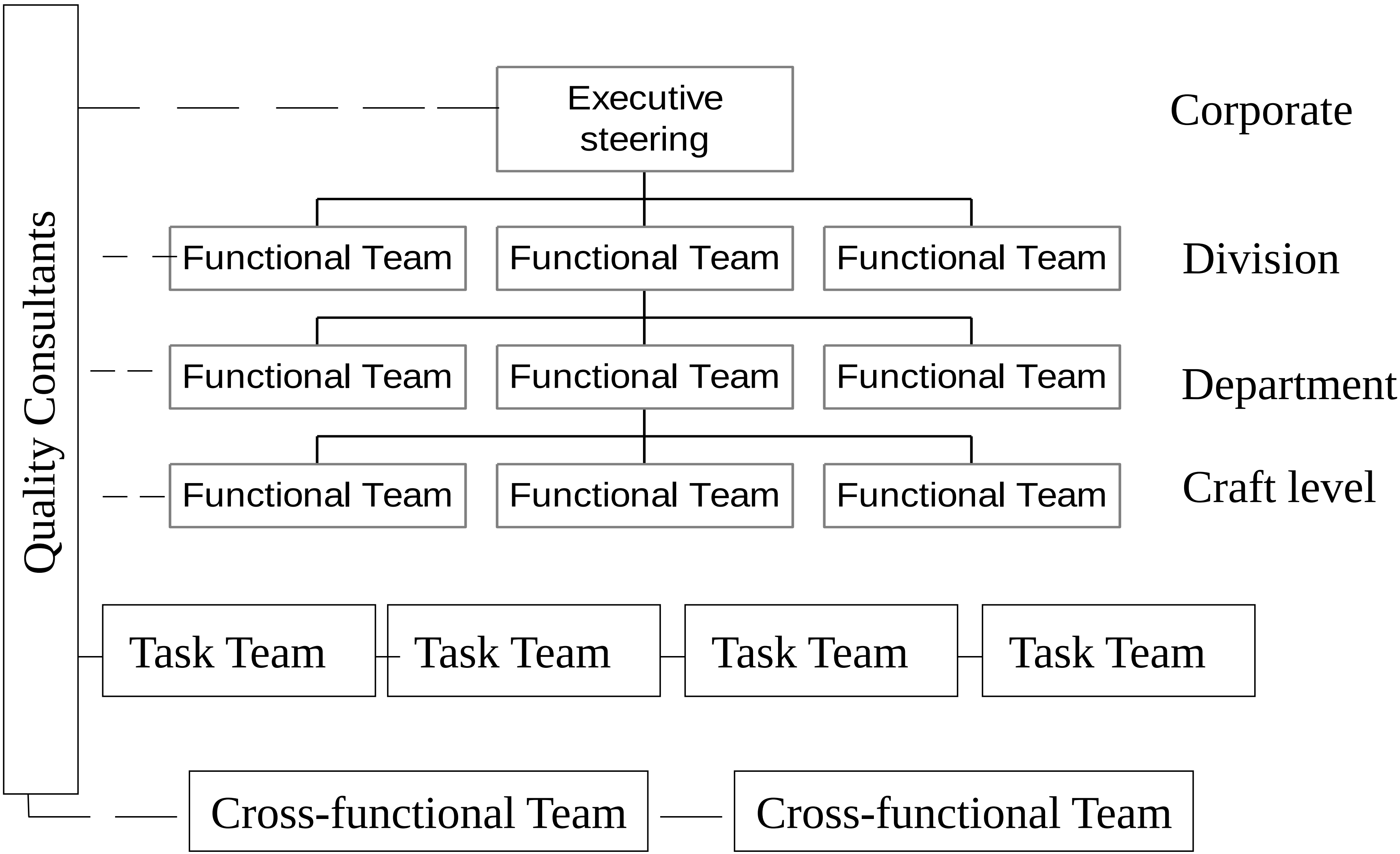
- Management commitment and leadership
- Training
- Teamwork
- Statistical methods
- Cost of quality
- Supplier involvement

TRAINING

- Training is very important part of TQM
- Should address both technical and humanistic aspects
- The training effort should be integrated with the work of individuals being trained
- Employees should apply newly learned skills to their jobs as quickly as possible

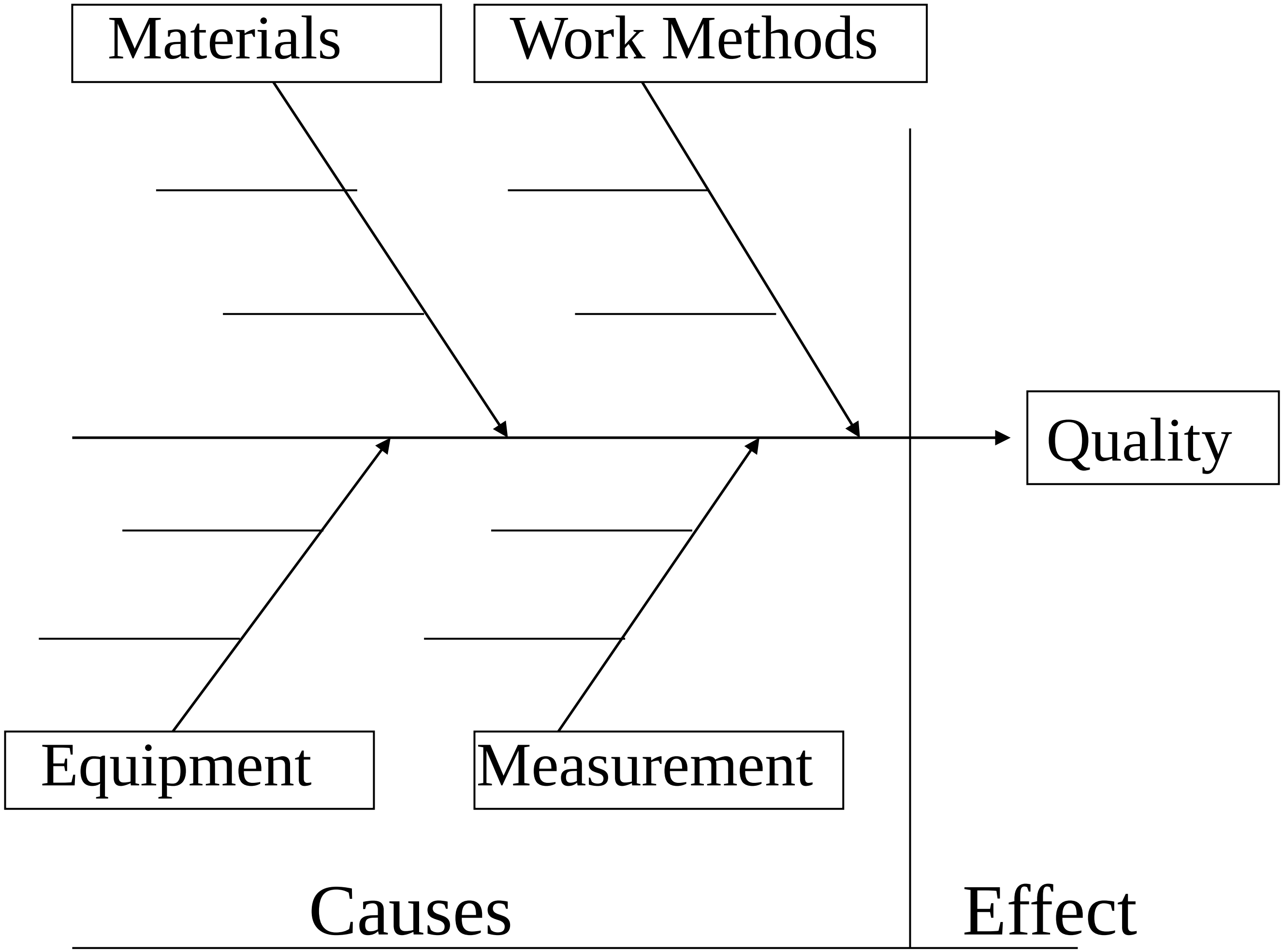
TRAINING TOPICS

- TQM philosophy
- TQM implementation plan
- Interpersonal skills
- Team development
- Leadership
- Statistical methods
- Problem solving
- Meeting rules and procedures
- Presentation skills

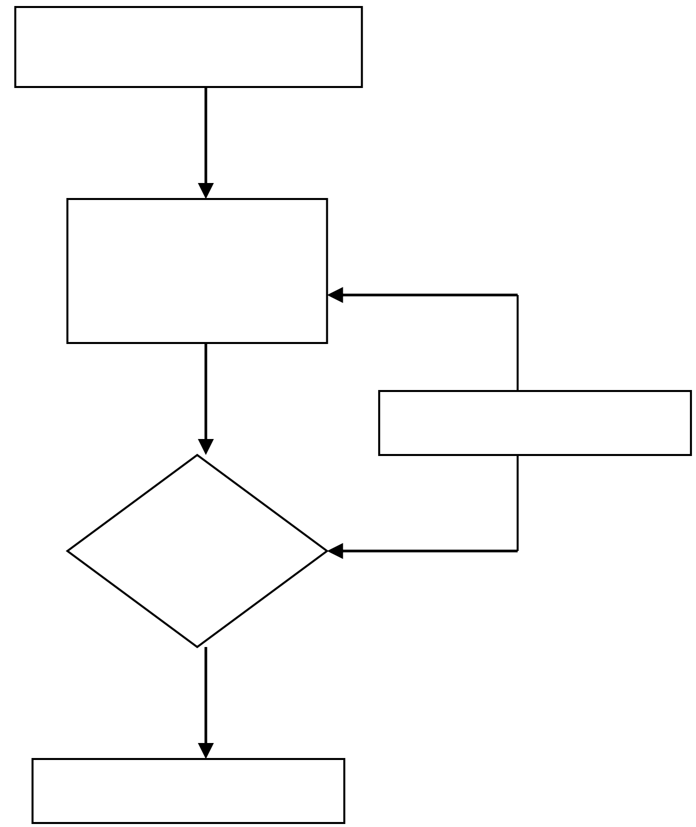

Quality Consultants
Executive steering
Corporate
Functional Team
Functional Team
Functional Team
Division
Functional Team
Functional Team
Functional Team
Department
Functional Team
Functional Team
Functional Team
Craft level
Task Team
Task Team
Task Team
Task Team
Cross-functional Team
Cross-functional Team

TOOLS FOR QUALITY

Cause - and - Effect Diagram

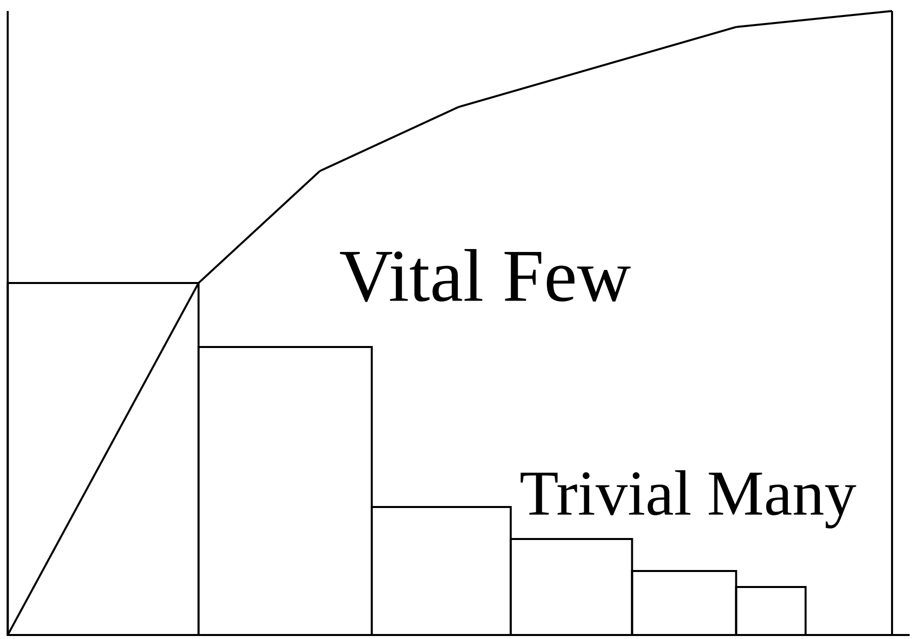


Flow Chart

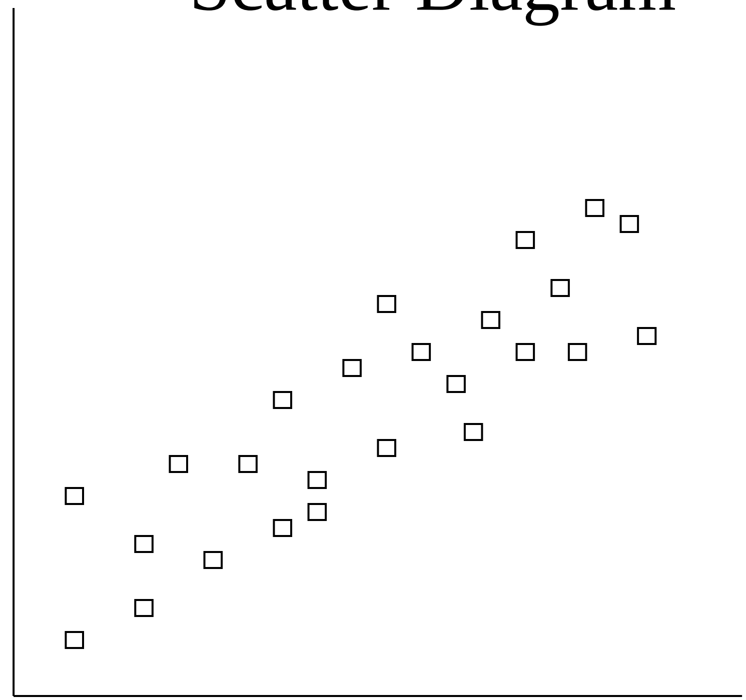


TOOLS FOR QUALITY

Pareto Chart

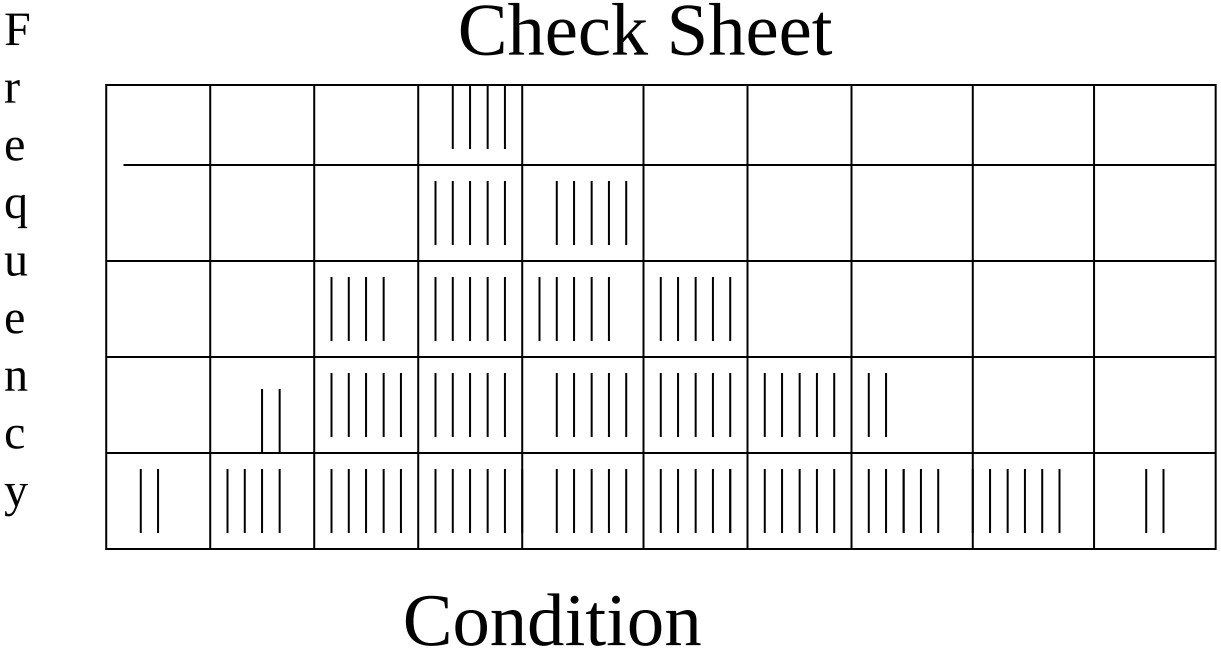


Scatter Diagram

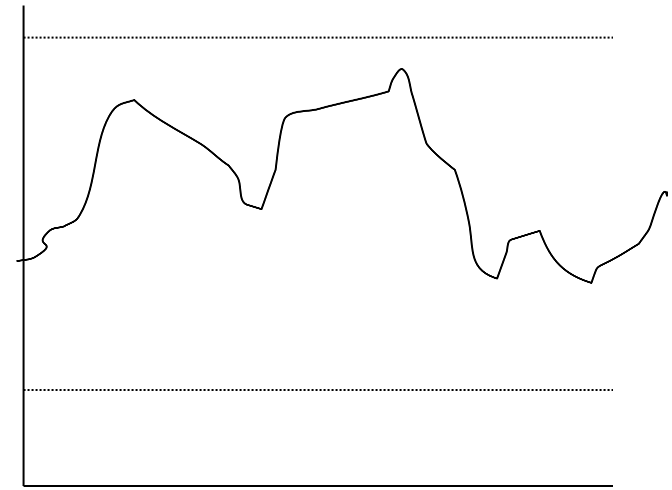


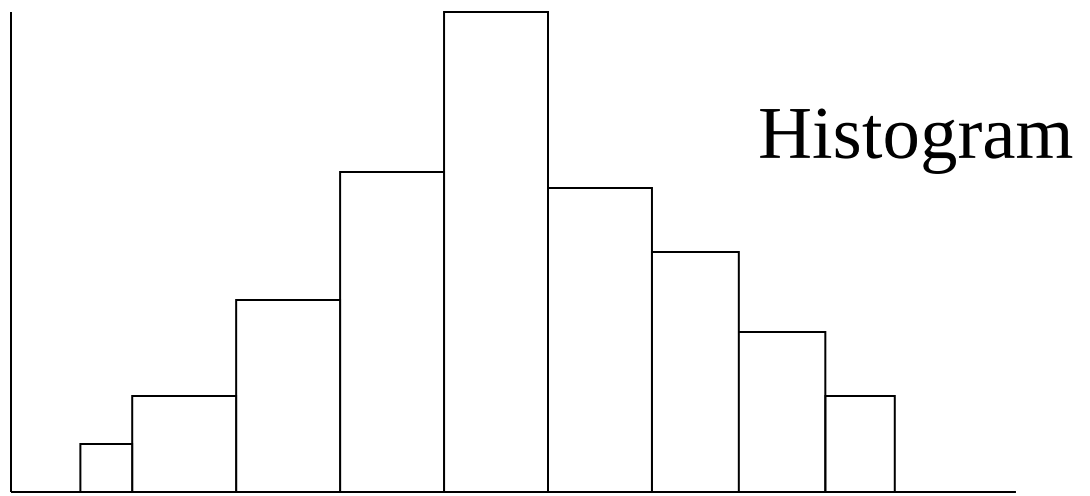
TOOLS FOR QUALITY

Check Sheet



Control Chart





Histogram

Quality Costs Breakdown

Quality Costs = Quality Management Costs
+Deviation Costs

Quality Management Costs = Prevention Costs
+ Appraisal Costs

- Prevention Costs
  - Those resulting from quality activities used to avoid deviations or errors.
  - eg., Design or Constructability reviews
- Appraisal Costs
  - costs incurred from quality activities used to detemine whether a product, process, or services conforms to established requiements.
  - eg., Inspection

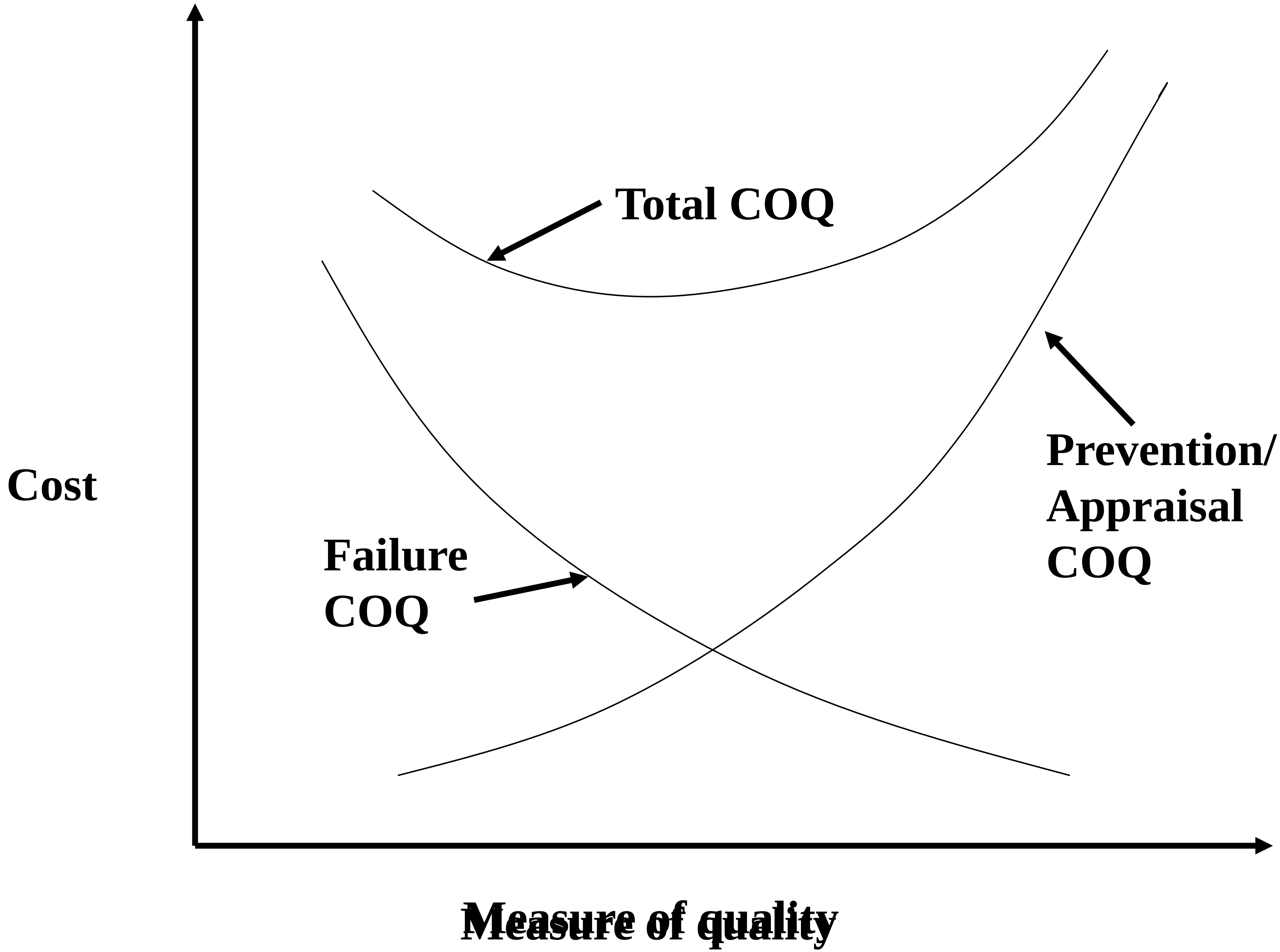
Deviation Costs

Deviation Costs = Internal Failure Costs + External Failure costs

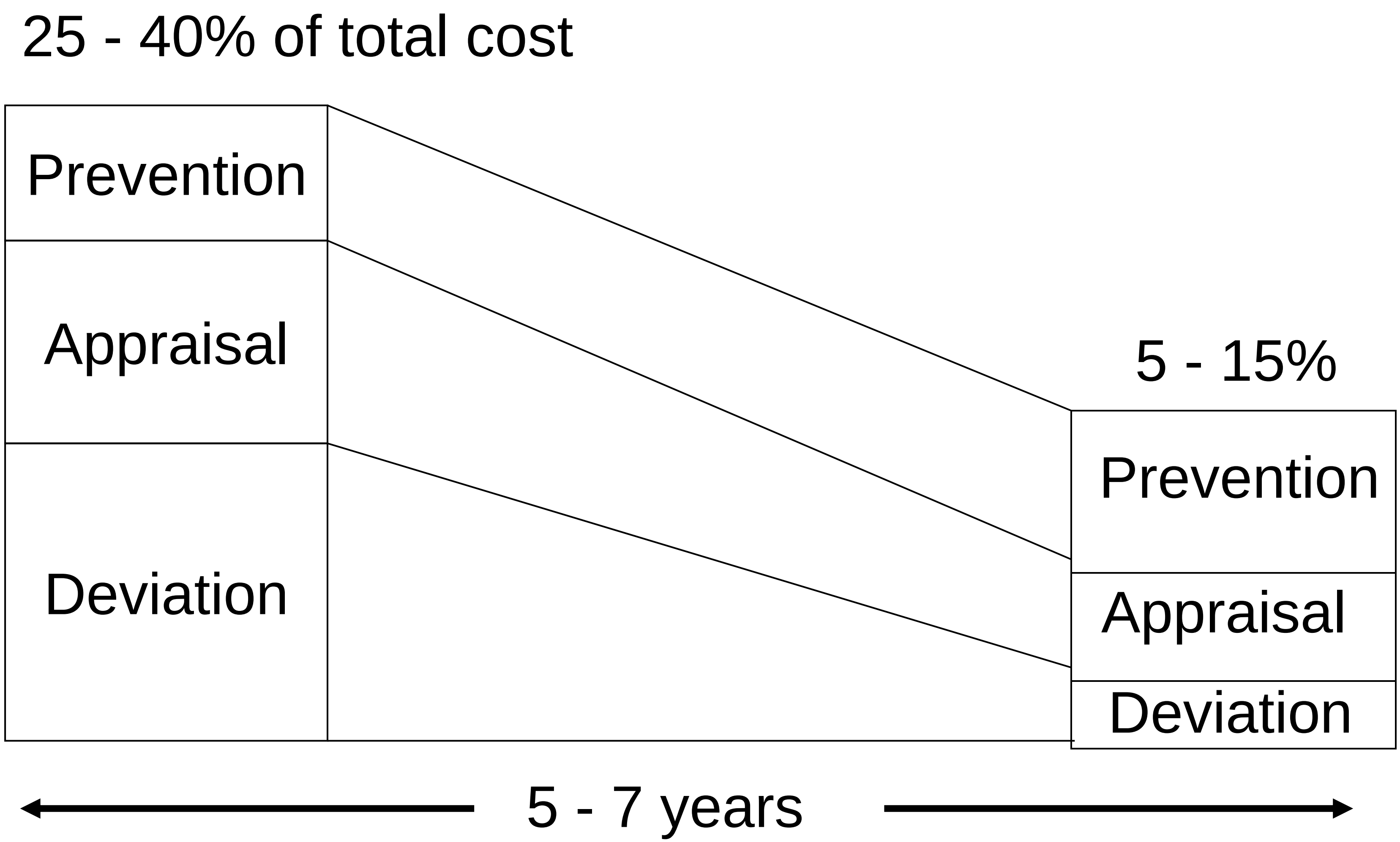
- Deviation Costs - those resulting from not meeting the requirements
- Internal Failure Costs - incurred on the project site

  - due to scrap, rework, failure analysis, reinspection, supplier error, or price reduction due to nonconformity
- External Failure Costs - incurred once the client has the project.

  - Costs of adjustments of complaints, Repair costs, Costs of handling and replacing rejected material, workmanship or equipment, Costs for correcting errors, Litigation Costs


Total COQ
Prevention/
Appraisal
COQ
Cost
Failure
COQ
Measure of quality

COST OF QUALITY



Type of Variation	Frequency of Occurrence	Characteristics	Action Needed	Responsibility
Common Cause	High (>90%)	Fault of the system	Fundamental system change	Management
Special Cause	Low (<10%)	Traceable to an assignable cause	Find the source and take preventive action	Local work force

QUALITY ASSURANCE

INTRODUCTION

- Quality assurance is a broad term encompassing the application of standards and procedures to ensure that a product or facility meets or exceeds a desired performance criteria
- Elements of quality:
  - **Quality characteristics**: properties that define the nature of a product for quality control purposes. Example- dimension, color, strength etc
  - **Quality of design**: designers often specify not only the desired characteristic that defines a product but also tolerances or ranges for acceptable variations from the standard
  - **Quality of conformance**: it is the degree to which the physical work produced conforms to the standards

ORGANIZATION FOR QUALITY ASSURANCE

- The quality assurance objectives of the various parties associated with a construction project differ and often conflict
  - The owner wants to maximize the quality of characteristics associated with the intended function of a project
  - The designer wants a level of quality that will assure satisfactory performance of the structure and be a credit to his professional reputation
  - A constructor on a fixed price contract will be interested mainly in satisfying the specifications at minimum direct cost
  - Increasingly, external regulatory agencies are setting quality standards characteristics that may not even be directly related to the primary function of the project, and often with little consideration for cost. Example- control on emissions to the environment

ORGANIZATION FOR QUALITY ASSURANCE

- Typical Organizations:
  - Resolving the conflicts in objectives between the various parties generally makes the organization and responsibilities of quality assurance system different from project planning and control system
  - For conformance oriented quality control, more attention is given to separating the responsibility for judging quality from those charged with carrying out the work
  - In traditional competitively bid public works, quality control is one of the primary responsibilities of the resident engineer and his inspectors
  - The relationship is shown in Figure 1
  - Even in design construct work, owner or governmental requirements often mandate that quality assurance be separated from production operations. Figure 16-5 shows a setup typical of a nuclear power plant

Fig. 1 Quality Control Organization in a Competitively Bid Public Works Project

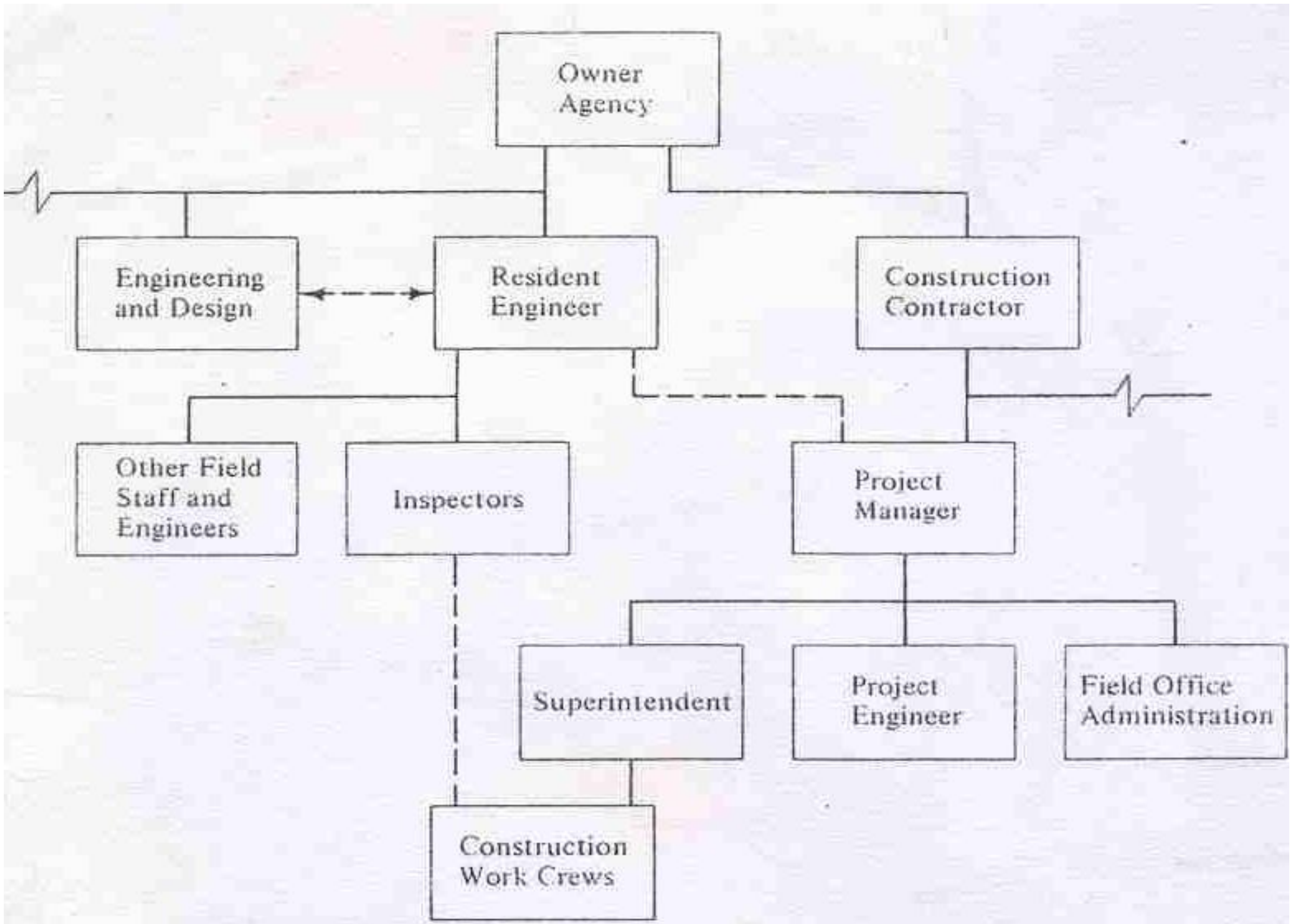
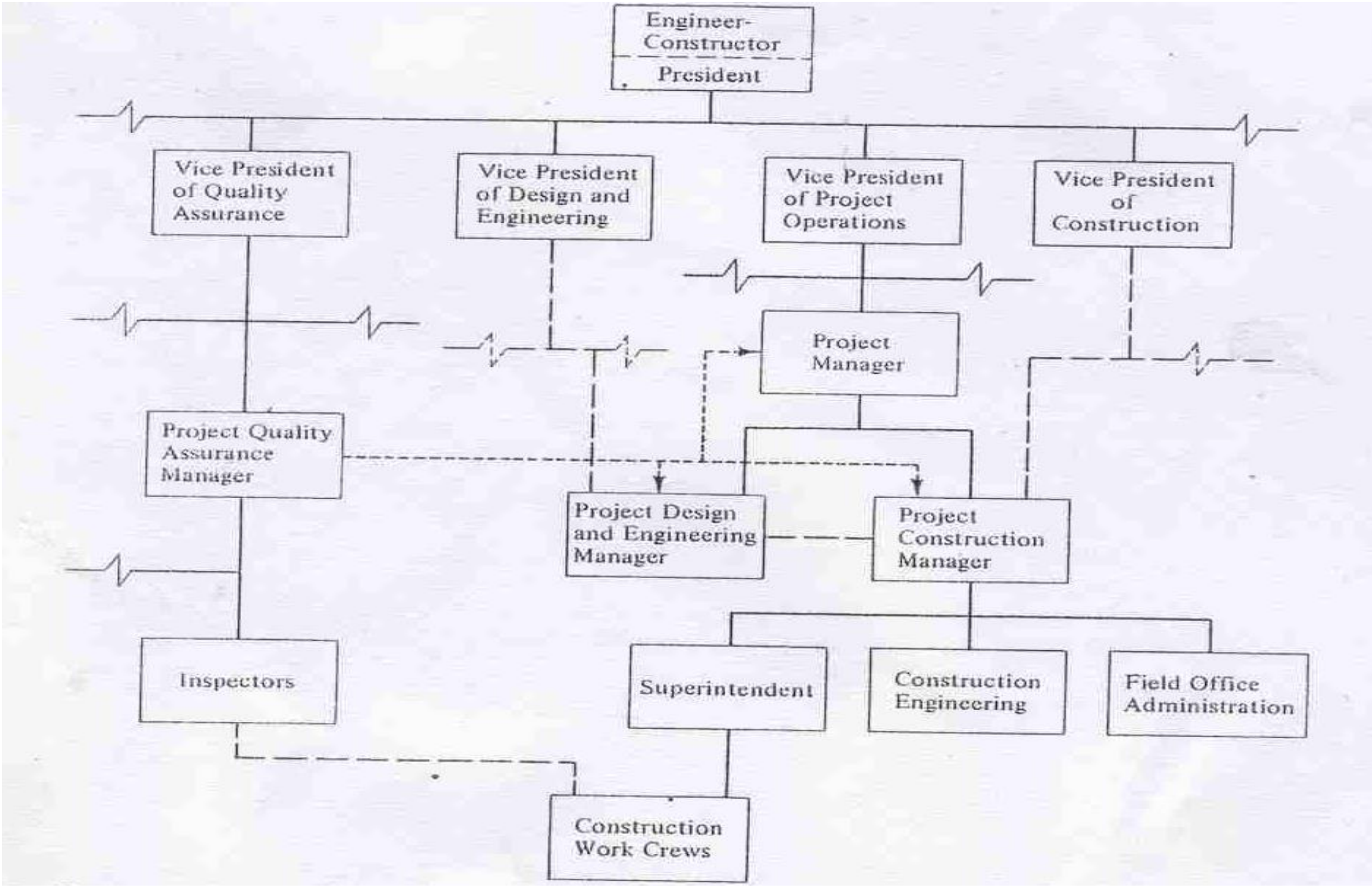


Fig 2. Quality Control Organization in Design-Construct Project

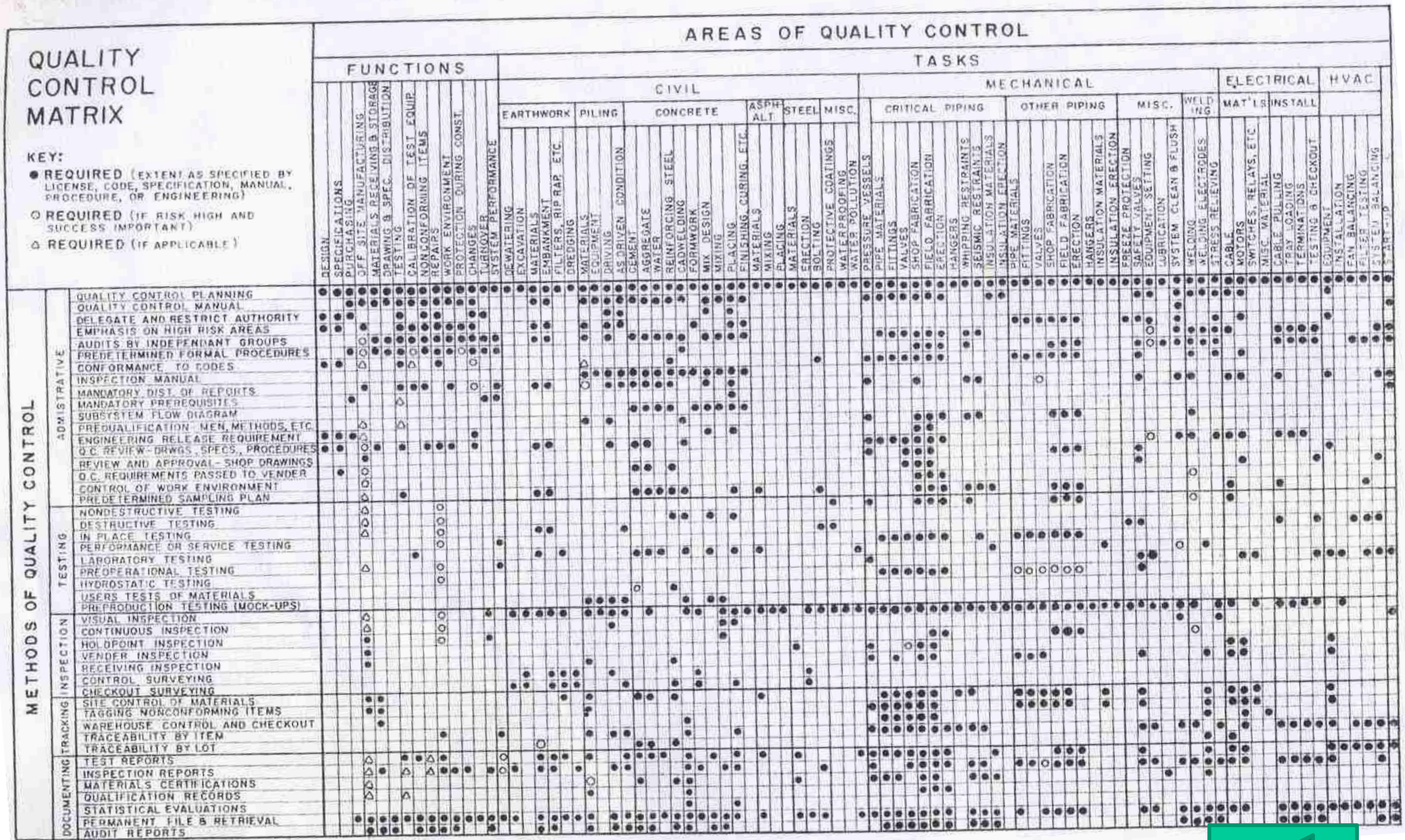


ORGANIZATION FOR QUALITY ASSURANCE

- The idea is to setup an autonomous, independent and objective group that is free to apply controls without fear of censure from higher levels in the operations organization
- In professional construction management, the professional construction manager provides objective third party input into determining the quality of design, as well as inspection services to control quality of conformance

- Tasks, Responsibilities and Procedures
  - Figure 3 uses a convenient matrix form to show the important relationships for quality assurance and summarizes the whole program
  - The preparation of such a chart can spotlight areas of conflict and omissions at the planning stage

Fig 3. Quality Control Matrix



Source: “System for Control of Construction Quality,” Journal of the Construction Division, ASCE, Vol. 98, No. CO1, March 1972

Journal of the Construction Division, ASCE, Vol. 98, No. CO1, Mar...

ORGANIZATION FOR QUALITY ASSURANCE

- Delegating Responsibility and Authority
  - In addition to the organization charts given in Figures 1 and 2, a linear responsibility chart, such as that presented in Figure 4, clearly shows the responsibilities and relationships among the key parties
  - Preparation of such a chart requires a thorough understanding of the project and the contractual relationships of the parties; it also clarifies ambiguities and identifies omissions and conflicts
- Implementation
  - Figure 5 gives a detailed example of how the procedures operate in the case of quality control on concrete construction

Fig 4. Linear Responsibility Chart (LRC) for Quality Control Organization

LINEAR RESPONSIBILITY CHART

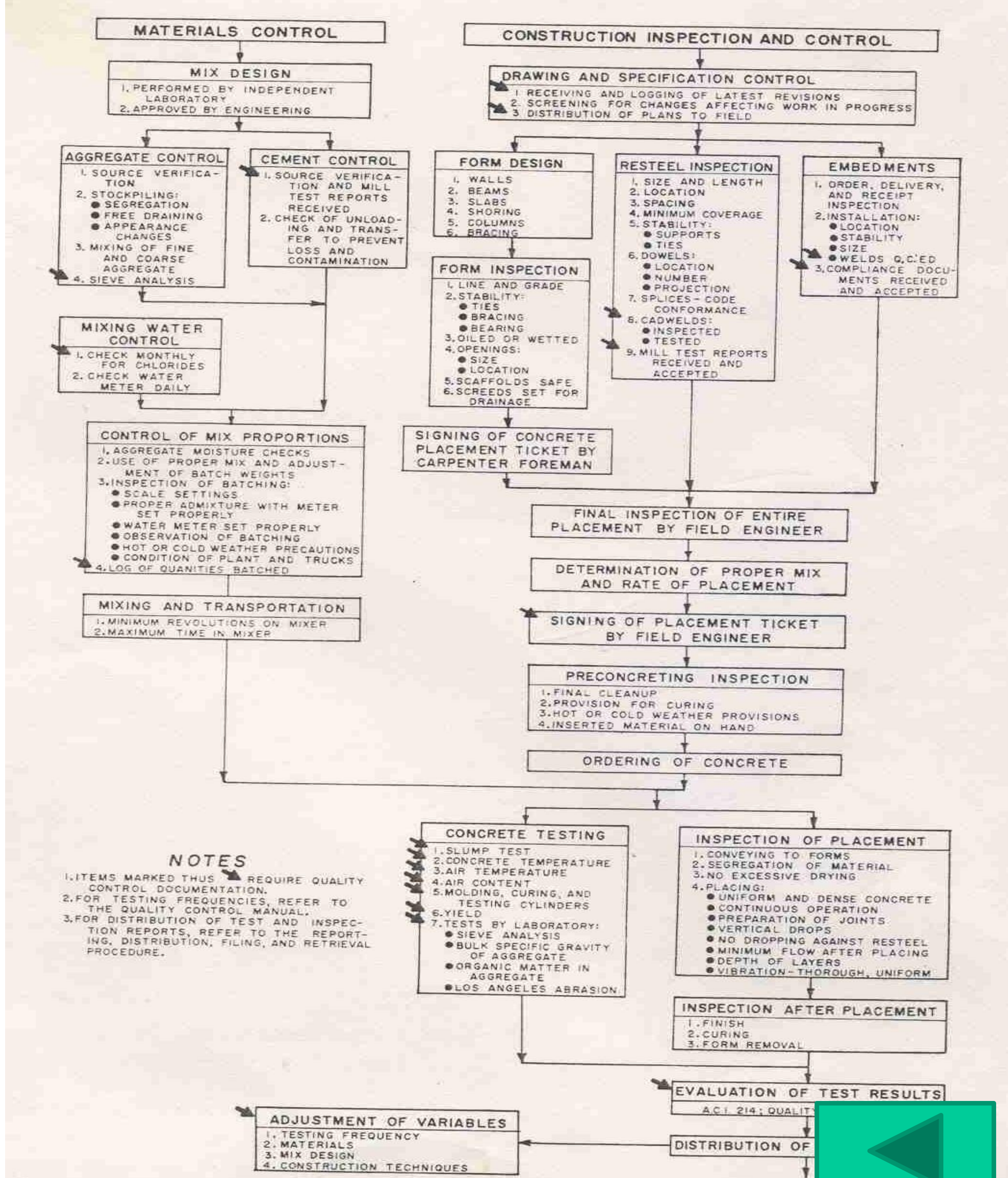
• KEY •

- PRIMARY RESPONSIBILITY ●
- JOINT RESPONSIBILITY ▲
- APPROVAL RESPONSIBILITY ■
- MUST BE CONSULTED ○
- MAY BE CONSULTED △
- AUDITS OR REVIEWS □

	OWNER				ENGINEER				CONSTRUCTOR							
	Manager of New Construction	Chief Engineer	Board of Review	Site Representative	Project Manager	Project Engineer	Responsible Design Engineer	Materials Engineer	Project Superintendent	Construction Superintendent	Site Q.C Supervisor	Site Q.A. Supervisor	Home Office Q.A. Chief	Purchasing Agent	Startup Engineer	Regulating Agency
Select quality objectives	▲				▲				▲							▲
Define activities affecting quality	■	△	□		●	△	△	△	○							□
Specify quality standards	■	△	□		■	●	△	△	○							□
Prepare quality control manual	■	△	□		■	○		△	○				●			□
Prepare quality control procedures		□			■					○	●		■			□
Prepare construction method procedures				□		□			■	●	○	□				□
Prepare welding and NDT procedures		□	□		■			●								□
Establish design criteria	■	●	□													□
Perform design	■	■	□		■	■	●									□
Define vendor quality control requirements		■			■	●	○	○						△		□
Prepare procurement documents	■	△	□		○								□	●		
Evaluate vendor quality capability	■	△	□		■	○			○					●		□
Inspect off site manufacturing		□	□			□	△					●		○		□
Control distribution of plans and specifications				□		○			○		●					□
Specify sampling plans		□	□			□	△		○				●			□
Train and qualify craftsmen				□					○	●	○	□				□
Train and qualify inspection personnel				□	□						●	□				□
Direct construction operations									○	●						
Inspect work in progress			□	□		□	△			△	●	□				□
Accept work in progress			□	□	△	□	△				●	□				□
Stop work in progress		△		□	○		△		○		●					□
Inspect materials upon receipt				□			△	△				●	□			□
Monitor and evaluate quality trends			□	□	□							●				□
Maintain file of quality control documents				□	□						●					□
Determine disposition of nonconforming items	■	△	□	△	●	△	△	△	○		○	○				□
Investigate failures		○	□	□	○	○	△	△	△	△	●	△	△			□
Release systems and components to operations				□					□		●				■	□
Conduct flushing and cleaning operations				■											●	□
Conduct preoperational testing			□	■											●	□
Accept completed plant as to quality	●	○	△													■

Source: "System for Control of Construction Quality," Journal of the Construction Division, ASCE, Vol. 98, No. CO1, March 1972

Fig 5. Flow Diagram, Quality Control of Concrete Construction



Source: “System for Control of Construction Quality,” Journal of the Construction Division, ASCE, Vol. 98, No. CO1, March 1972

REQUEST FOR INSPECTION

RFI No:

Package ______ Section ______ Contractor ______

☐ Initial RFI ☐ Resubmittals Previous RFI Nos./ Date

Material ______ Source ______ Quantity ______

Activity	Chainage	BOQ Item	Activity Description	Schedule		Inspection Results/Remarks
				Date	Time	
						☐ APPROVED ☐ REJECTED * ☐ NOT READY * * See the comments below

Requested by Contractor

Date/Time:

Received by Res. Engr.

Date/Time:

Inspected by

Date/Time:

Comments :

Request to begin activity is: APPROVED / REJECTED

Resident Engineer
Date/Time

Contractor
Date/Time

Note:

1. Contractor to submit request minimum of 24 hours in advance of work requirements.
2. Consultants to return approved or disapproved original and one copy to the contractor before work proceeds

JMM JAMES M. MONTGOMERY, CONSULTING ENGINEERS, INC.

NON-COMPLIANCE NOTICE

To L&M Construction Co., Inc.
1234 Fontana Street
Marysville, CA 90000
Att: Paul T. Marikian

Date	15 May 1979
Time	09:45 A.M./P.M.
Inspector	Jackson
	APC Labs, Inc.

Project Title Interceptor & Collector Sewers Project No. F-345

Contractor Jorgensen & Sons Contract No. G07-5674

You are hereby notified that ☒ tests ☒ inspection indicates that the trench backfill and compaction between Sta 67+45 and 85+20 does not conform to the specifications requirements. The specification violated is Section 02450 Article/Paragraph 3.4(A) Under the provisions of the contract specifications, the requirements are sand be placed in pipe zone and after consolidation backfill be placed in lifts not exceeding 12 inches in depth and compacted to 90 percent per ASTM D 1557-70.

Noncomplying work may be required to be removed and replaced at no cost to the Owner.

It shall be your responsibility to determine the corrective action necessary, and to determine whether you wish to discontinue operations until additional investigations by the Owner or Engineer confirm or refute the initial findings.

Data used are fictitious for illustration only

R. G. Barnes
Resident Project Representative

Noncompliance notice was received by Contractor on May 15, 1979
date

DISTRIBUTION: 1. Contractor
2. Proj. Mgr.
3. Field Office
4. File
5. Owner

By Harold Martinson

Title Project Manager
L. & M. Construction Co., Inc

Purpose: Written notice of deficiency and a demand for corrective action.

Prepared by: Resident Project Representative; Receipt of notice acknowledged by the contractor's on-site manager.

Directed to: Prime contractor (including for deficiencies of subcontractors).

Copies to: Project Manager; Resident Project Representative; contractor; owner; project master file.

Comments: Description should contain accurate locations and spec references. The document may figure strongly in later claims by the contractor.

CONCRETE QUALITY CONTROL

- OSHA Regulations – Concrete
  - Equipment and materials used in concrete construction must meet the requirements of ANSI A10.9, "Safety requirements for concrete construction and masonry work"
- Concrete formwork
  - From the viewpoint of inspection, two factors are primary: safety and finish
  - During the placing of the concrete and development of initial set, safety is of the greatest importance, and deserves total emphasis
  - After removal of the formwork, finishing has to be taken care of
  - Concrete Formwork Checklist:
    - Check location of forms
    - Check interior condition of forms for soundness and proper surface finish

CONCRETE QUALITY CONTROL

- Check form supports for conformance with form drawings and basic soundness
- Check shoring and bracing in accordance with the shoring list
- Check for grade and soundness in formwork made of reused lumber
- Form should be wet down prior to use to swell them
- Forms should be cleaned out before pore
- Where form ties are used, check installation and number
- Metal forms should be oiled rather than wetted, but surplus oil should not be on the formwork
- Steel reinforcement is to be placed within forms in accordance with rebar drawings. Count the numbers and check sizes
- Reinforcing bars should be supported by chairs or is permitted by specifications, concrete blocks to hold them the proper distance away from the outside face of the concrete
- Where the final surface will be exposed, specifications usually call for plastic, galvanized tip, or stainless steel chairs or accessories to avoid rust marks

CONCRETE QUALITY CONTROL

- Cast-in-place Concrete
  - Concrete placing checklist
    - Check forms and rebar
    - Be sure that concrete is in accordance with design mix, certified id required
    - Inspect at batching plant if required
    - Check time of receipts for ready mix, amount of water added, if permitted, and mixing time
    - Be sure trucks do not overstay maximum waiting period before placing of concrete, usually less than an hour
    - Be sure that placement is in accordance with specifications
    - Concrete should be dropped only the allowable distance, and should not bounce across reinforcing bars or other obstructions causing segregation

CONCRETE QUALITY CONTROL

- Be sure that appropriate slump and air content tests are taken as required
- Concrete should be vibrated as it is placed, and vibrators should not hit the reinforcing bars or the forms. Concrete should be worked either with a float or a troweled finish, as required by the specifications
- Curing compound should be placed on finished concrete, unless it is under a floor to be tile covered
- Concrete should not be placed after initial set has been reached, and should not be remixed after initial set
- Screed boards are often utilized to assure the proper grade of finished concrete
- Be sure that test cylinders are taken and handled carefully. They should be stored under job conditions

CONCRETE QUALITY CONTROL

- Concrete Test Checklist
  - Check mix versus approved mix design
  - Check truck time from start of mixing
  - Take slump test if required
  - Take air test for air entrained concrete
  - Take concrete cylinders, and:
    - Mix and fill cylinders
    - Rod carefully
    - Strike off and tap to close voids
    - Cure under job conditions
  - Check test results at 3 or 7 days and confirm at 28 days